

CUSTOMER SHOPPING BEHAVIOR ANALYSIS

Data Analytics Project Report

Submitted By:

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Tools & Technologies Used

- Python
- Pandas
- NumPy
- Jupyter Notebook
- PostgreSQL
- SQL
- SQLAlchemy
- Psycopg2
- Power BI

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1. Introduction

In today's competitive retail market, understanding customer purchasing behavior is essential for making informed business decisions. This project focuses on analyzing customer shopping data using Python, PostgreSQL, SQL, and Power BI to identify trends, customer preferences, purchasing patterns, and business opportunities.

The project demonstrates a complete data analytics workflow, beginning with raw data preprocessing, followed by SQL-based business analysis and interactive dashboard development using Power BI.

2. Project Objectives

The primary objectives of this project are:

- Clean and preprocess customer shopping data.
 - Handle missing values efficiently.
 - Create useful business features through feature engineering.
 - Store the cleaned data inside PostgreSQL.
 - Perform SQL analysis to answer business questions.
 - Develop an interactive Power BI dashboard.
 - Generate business insights for decision-making.
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3. Dataset Overview

The dataset contains customer shopping information collected from an online retail platform.

Dataset Summary

- Total Records: **3900**
- Total Columns: **18**
- Data Type: Mixed (Numerical & Categorical)

Important Columns

- Customer ID
- Age

- Gender
- Item Purchased
- Category
- Purchase Amount
- Review Rating
- Discount Applied
- Subscription Status
- Shipping Type
- Previous Purchases
- Payment Method
- Frequency of Purchases

Dataset Preview

```
[1]: import pandas as pd

df = pd.read_csv('customer_shopping_behavior.csv')
```

```
[2]: df.head()
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	F
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	

```
[3]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   Customer ID           3900 non-null   int64
 1   Age                   3900 non-null   int64
 2   Gender                 3900 non-null   object
 3   Item Purchased         3900 non-null   object
 4   Category               3900 non-null   object
 5   Purchase Amount (USD)  3900 non-null   int64
 6   Location                3900 non-null   object
 7   Size                   3900 non-null   object
 8   Color                  3900 non-null   object
 9   Season                 3900 non-null   object
10   Review Rating          3863 non-null   float64
11   Subscription Status     3900 non-null   object
12   Shipping Type           3900 non-null   object
13   Discount Applied        3900 non-null   object
14   Promo Code Used         3900 non-null   object
15   Previous Purchases      3900 non-null   int64
16   Payment Method          3900 non-null   object
17   Frequency of Purchases  3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

```
[4]: df.describe(include='all')
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	F
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900	3900	
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2	2	
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No	No	
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223	2223	
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN	NaN	
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN	NaN	
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN	NaN	
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN	NaN	
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN	NaN	

Observations

- Dataset contains customer demographic details.
- Includes purchasing behaviour.
- Contains customer subscription information.
- Contains customer review ratings.
- Useful for customer segmentation and revenue analysis.

4. Technologies Used

Technology	Purpose
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Python	Data Analysis
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Pandas	Data Cleaning
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NumPy	Numerical Operations
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SQLAlchemy	Database Connection
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Psycopg2	PostgreSQL Connector
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PostgreSQL	Database Storage
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SQL	Business Analysis
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Power BI	Dashboard & Visualization
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5. Data Cleaning & Preprocessing

5.1 Loading Dataset

The dataset was imported into Jupyter Notebook using Pandas.

5.2 Dataset Information

The dataset structure was inspected using:

- `head()`
- `info()`
- `describe()`

```
[5]: df.isnull().sum()

[5]: Customer ID      0
     Age            0
     Gender         0
     Item Purchased  0
     Category        0
     Purchase Amount (USD) 0
     Location        0
     Size           0
     Color          0
     Season         0
     Review Rating   37
     Subscription Status 0
     Shipping Type   0
     Discount Applied 0
     Promo Code Used 0
     Previous Purchases 0
     Payment Method  0
     Frequency of Purchases 0
     dtype: int64

[6]: df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x: x.fillna(x.median()))

[7]: df.isnull().sum()

[7]: Customer ID      0
     Age            0
     Gender         0
     Item Purchased  0
     Category        0
     Purchase Amount (USD) 0
     Location        0
     Size           0
     Color          0
     Season         0
     Review Rating   0
     Subscription Status 0
     Shipping Type   0
     Discount Applied 0
     Promo Code Used 0
     Previous Purchases 0
     Payment Method  0
     Frequency of Purchases 0
     dtype: int64

[8]: df.columns = df.columns.str.lower()
     df.columns = df.columns.str.replace(' ','_')
     df = df.rename(columns = {'purchase_amount_(usd)' : 'purchase_amount'})

[9]: df.columns

[9]: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
        'purchase_amount', 'location', 'size', 'color', 'season',
        'review_rating', 'subscription_status', 'shipping_type',
        'discount_applied', 'promo_code_used', 'previous_purchases',
        'payment_method', 'frequency_of_purchases'],
        dtype='object')

[10]: # create a age group
     labels = ['Young Adult','Adult','Middle-aged','senior']
     df['age_group'] = pd.qcut(df['age'], q=4, labels = labels)

[11]: df[['age','age_group']].head(10)

[11]:   age  age_group
0    55  Middle-aged
1    19  Young Adult
2    50  Middle-aged
3    21  Young Adult
```

Observations

- Dataset contains 3900 rows.
- No duplicate customer IDs.
- Numerical and categorical features present.
- Review Rating contained missing values.

5.3 Missing Value Analysis

The missing values were identified using:

```
df.isnull().sum()
```

Observation

Only the **Review Rating** column contained missing values.

5.4 Missing Value Treatment

Instead of deleting records, missing review ratings were replaced using the median review rating of each product category.

Why Median?

- Prevents data loss.
- Reduces effect of outliers.
- Maintains category consistency.

[Insert Screenshot]

5.5 Standardizing Column Names

Column names were converted to lowercase.

Spaces were replaced using underscores.

This improves SQL compatibility.

[Insert Screenshot]

5.6 Removing Redundant Columns

The project identified that:

- Promo Code Used
- Discount Applied

contained identical information.

Therefore,

Promo Code Used was removed.

Benefits:

- Reduced redundancy.

- Improved database normalization.
-

6. Feature Engineering

6.1 Age Group Creation

Customers were divided into four age groups:

Age	Group
18–30	Young Adult
31–45	Adult
46–60	Middle Aged
60+	Senior

This enables demographic analysis.

[Insert Screenshot]

6.2 Purchase Frequency Days

Purchase frequency was converted into numerical values.

Example:

- Weekly → 7
- Bi-weekly → 14
- Monthly → 30

Benefits:

- Easier statistical analysis.
 - Better dashboard visualization.
-

7. PostgreSQL Integration

The cleaned dataset was successfully exported into PostgreSQL using SQLAlchemy.

Steps

- Database connection established.

- Data uploaded.
- SQL queries executed.

```

7 27 Young Adult
8 26 Young Adult
9 57 Middle-aged

[12]: #create columns purchase_frequency_days

frequency_mapping = {
    'Fortnightly' : 14,
    'Weekly' : 7,
    'Monthly' : 30,
    'Quarterly' : 90,
    'Bi-Weekly' : 14,
    'Annually' : 365,
    'Every 3 month' : 90
}

df['purchase_frequency_days'] = df['frequency_of_purchases'].map(frequency_mapping)

[13]: df[['purchase_frequency_days', 'frequency_of_purchases']].head(10)

[13]:   purchase frequency days  frequency of purchases
0                14.0      Fortnightly
1                14.0      Fortnightly
2                 7.0        Weekly
3                 7.0        Weekly
4               365.0        Annually
5                 7.0        Weekly
6                NaN        Quarterly
7                 7.0        Weekly
8               365.0        Annually
9                NaN        Quarterly

[14]: (df['discount_applied'] == df['promo_code_used']).all()

[14]: np.True_

[15]: df = df.drop('promo_code_used', axis=1)

[16]: df.columns

[16]: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
        'purchase_amount', 'location', 'size', 'color', 'season',
        'review_rating', 'subscription_status', 'shipping_type',
        'discount_applied', 'previous_purchases', 'payment_method',
        'frequency_of_purchases', 'age_group', 'purchase_frequency_days'],
        dtype='object')

[17]: pip install psycpg2-binary sqlalchemy

Defaulting to user installation because normal site-packages is not writeable
Collecting psycpg2-binary
  Downloading psycpg2_binary-2.9.12-cp313-cp313-win_amd64.whl.metadata (5.1 kB)
Requirement already satisfied: sqlalchemy in C:\ProgramData\anaconda3\Lib\site-packages (2.0.43)
Requirement already satisfied: greenlet>=1 in C:\ProgramData\anaconda3\Lib\site-packages (from sqlalchemy) (3.2.4)
Requirement already satisfied: typing-extensions>=4.6.0 in C:\Users\troub\AppData\Roaming\Python\Python313\site-packages (from sqlalchemy) (4.15.0)
Downloading psycpg2_binary-2.9.12-cp313-cp313-win_amd64.whl (2.8 MB)
----- 0.0/2.8 MB ? eta -?:--
----- 0.5/2.8 MB 4.1 MB/s eta 0:00:01
----- 1.3/2.8 MB 3.8 MB/s eta 0:00:01
----- 2.1/2.8 MB 3.9 MB/s eta 0:00:01
----- 2.8/2.8 MB 3.7 MB/s 0:00:00
Installing collected packages: psycpg2-binary
Successfully installed psycpg2-binary-2.9.12
NOTE: you may need to restart the kernel to use updated packages

```

8. SQL Business Analysis

Q1 Revenue Generated by Gender

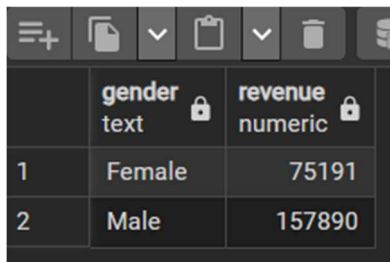
Objective

Compare revenue generated by male and female customers.

SQL Query

```
select gender, SUM(purchase_amount) as revenue  
from customers_data  
group by gender
```

Output



A screenshot of a database query result table. The table has two columns: 'gender' (text) and 'revenue' (numeric). The data is as follows:

	gender text	revenue numeric
1	Female	75191
2	Male	157890

Interpretation

- Compare total revenue generated by each gender.
- Identify the higher revenue-generating customer segment.

Business Value

Helps marketing teams target high-value customer groups.

Q2 Customers Using Discounts but Spending Above Average

Objective

SQL Query

Identify customers who used discounts but still spent more than the average purchase amount.

```
select customer_id, purchase_amount
```

```
from customers_data
```

```
where discount_applied = 'Yes' and purchase_amount >= (select AVG(purchase_amount)  
from customers_data)
```

Output

Data Output Messages Notifications			
≡+ 📄 ▼ 📋 ▼ 🗑️ 🗄️ ⬇️ 📈 SQL			
	customer_id bigint		purchase_amount bigint
1	2		64
2	3		73
3	4		90
4	7		85
5	9		97
6	12		68
7	13		72
8	16		81
9	20		90
10	22		62
11	24		88
12	29		94
13	32		79

Interpretation

These customers remain high-value even while using promotional offers.

Business Value

Ideal candidates for loyalty programs and premium memberships.

Q3 Highest Rated Products

Objective

Find products with the highest average customer ratings.

Data Output Messages Notifications		
	item_purchased text	Average product rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

Interpretation

Highly rated products indicate strong customer satisfaction.

Business Value

Increase marketing exposure for these products.

Q4 Standard vs Express Shipping

Objective

Compare average purchase amount between shipping methods.

Data Output Messages Notifications		
	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

Interpretation

Shipping preference influences customer spending.

Business Value

Optimize shipping strategies and pricing.

Q5 Subscription Analysis

Objective

Compare subscriber and non-subscriber spending.

Data Output Messages Notifications				
	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

Interpretation

Determine whether subscriptions improve customer lifetime value.

Business Value

Helps evaluate subscription effectiveness.

Q6 Discount Rate by Product

Objective

Find products with the highest percentage of discounted purchases.

Data Output Messages Notifications		
	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

Interpretation

Some products rely heavily on discounts.

Business Value

Optimize promotional campaigns.

Q7 Customer Segmentation

Segments

- New

- Returning
- Loyal

Data Output		Messages	Notifications
			
			
			
	customer_segment text	Number of Customers bigint	
1	New	83	
2	Loyal	3116	
3	Returning	701	

Business Value

Allows personalized marketing strategies.

Q8 Top Products in Each Category

Objective

Identify the three best-selling products within every category.

Data Output

Messages

Notifications

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

Business Value

Improves inventory management.

Q9 Repeat Buyers vs Subscription

Objective

Check whether repeat buyers are subscribers.

Data Output		Messages	Notifications
	subscription_status text	repeated_buyers bigint	
1	No	2518	
2	Yes	958	

Interpretation

Shows customer loyalty behaviour.

Business Value

Target repeat buyers with subscription campaigns.

Q10 Revenue by Age Group

Objective

Analyze revenue generated by each age group.

Data Output		Messages	Notifications
	age_group text	total_revenue numeric	
1	Young Adult	62143	
2	Middle-aged	59197	
3	Adult	55978	
4	senior	55763	

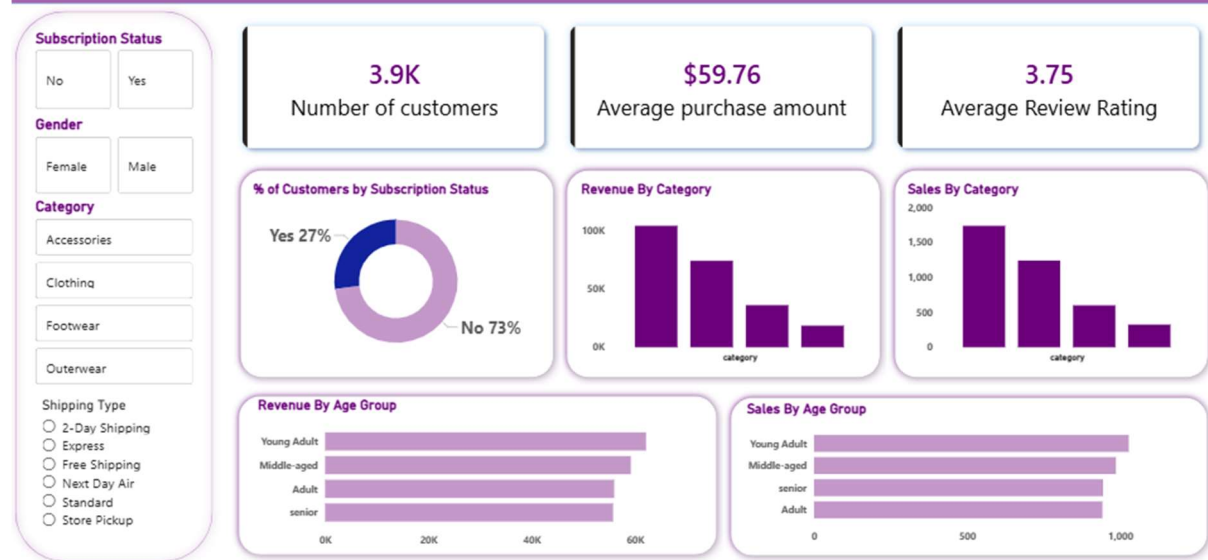
Business Value

Improves age-based marketing strategies.

9. Power BI Dashboard

ent theme (Default)

Customer Shopping Behavior Dashboard



Dashboard Components

KPI Cards

- Total Customers
- Average Purchase Amount
- Average Review Rating

Charts

- Revenue by Category
- Sales by Category
- Revenue by Age Group
- Sales by Age Group
- Subscription Distribution

Filters

- Gender
- Subscription Status
- Category
- Shipping Type

Dashboard Insights

- Interactive filtering enables dynamic analysis.
 - Revenue trends can be explored by demographic attributes.
 - Category performance becomes immediately visible.
 - Customer behaviour can be analyzed across multiple dimensions.
-

10. Business Insights

1. **Revenue distribution by gender** highlights which customer group contributes more to overall sales, helping tailor marketing strategies.
2. **Customers who use discounts but still spend above the average purchase amount** represent valuable customers who respond well to promotions without significantly reducing basket size.
3. **Products with the highest review ratings** demonstrate strong customer satisfaction and should receive greater promotional visibility.
4. **Shipping preferences influence purchasing behavior**, indicating that delivery options can affect order values and customer experience.
5. **Subscriber spending patterns** help determine whether subscription programs effectively increase customer lifetime value.
6. **Products with high discount rates** may rely heavily on promotional pricing, suggesting the need to evaluate long-term profitability.
7. **Customer segmentation into New, Returning, and Loyal groups** enables personalized engagement strategies and targeted retention efforts.
8. **Top-performing products within each category** provide valuable insights for inventory planning and merchandising decisions.
9. **Repeat buyers who are not subscribers** present an opportunity to increase recurring revenue through targeted subscription offers.
10. **Revenue contribution by age group** helps identify the most valuable customer demographics, allowing businesses to optimize marketing campaigns and product offerings.
11. **Interactive Power BI dashboards** allow stakeholders to monitor KPIs, filter customer segments, and make data-driven decisions in real time.

12. Combining Python, SQL, and Power BI creates a complete analytics pipeline that transforms raw customer data into actionable business intelligence.

11. Recommendations

- Increase investment in high-performing product categories.
 - Expand loyalty and subscription programs for repeat customers.
 - Use targeted discounts instead of blanket promotions.
 - Promote top-rated products more aggressively.
 - Personalize marketing campaigns by age group and gender.
 - Monitor subscription conversion among loyal customers.
 - Optimize shipping options based on spending behavior.
 - Use customer segmentation for personalized offers.
 - Continuously monitor customer satisfaction through review ratings.
 - Extend the project with predictive analytics to forecast future sales and customer behavior.
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12. Conclusion

This project successfully demonstrated a complete end-to-end data analytics workflow by integrating Python, PostgreSQL, SQL, and Power BI. The dataset was cleaned, transformed, and enriched through feature engineering before being stored in a relational database. Business-focused SQL queries provided meaningful insights into customer purchasing behavior, while the interactive Power BI dashboard enabled effective visualization of key performance indicators and trends.

The analysis highlights how data-driven decision-making can help businesses better understand customer demographics, purchasing patterns, product performance, and subscription behavior. By applying the recommendations identified in this report, organizations can improve customer retention, optimize marketing strategies, enhance inventory planning, and ultimately increase revenue. This project also serves as a strong foundation for future enhancements such as predictive analytics, customer segmentation using machine learning, and demand forecasting.